

Harder but not Better? Measuring the Impact of LLM-Generated Distractors in MCQA



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LLM Distractors Look Good at First ... But are they truly “harder”?

We determine the difficulty and quality of LLM vs. human distractors in Multiple-Choice Questions (MCQA).

What process in the plant is responsible for the phenomenon of guttation?

A. Guttation is the outcome of rapid cellular respiration in plant tissues.

B. Guttation is a result of photosynthesis.

C. Guttation is caused by external weather conditions.

D. Guttation is explained by the effect of root pressure.

E. Guttation is a byproduct of the plant’s defense mechanisms.

F. Guttation is caused by the plant’s response to light exposure.

G. Guttation happens as a result of osmotic pressure in the atmosphere.

H. Guttation is a consequence of nutrient transport within the plant.

I. Guttation occurs due to capillary action in leaves.

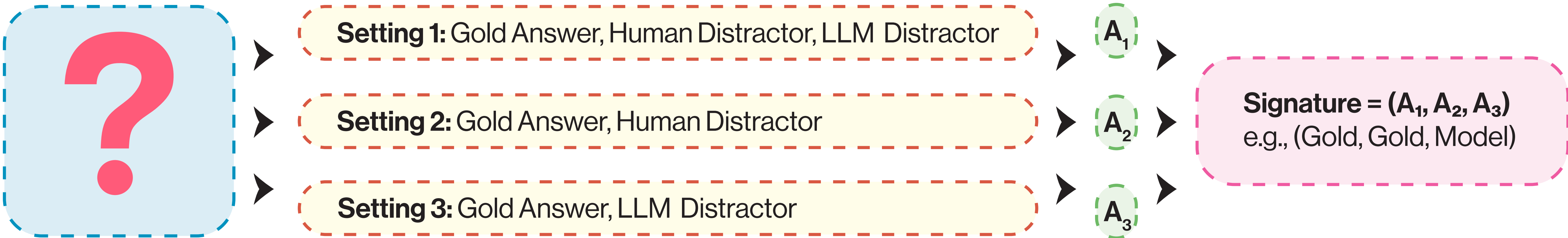
J. Guttation is due to the process of transpiration.

Gold answer

Human-generated distractor

LLM-generated distractor

Experiment



> A model (GPT 4.1) is tested against each question from MMLU-Pro in three configurations to see how its answer changes. MMLU-Pro is an augmented form of MMLU, with 6 additional synthetic distractors.

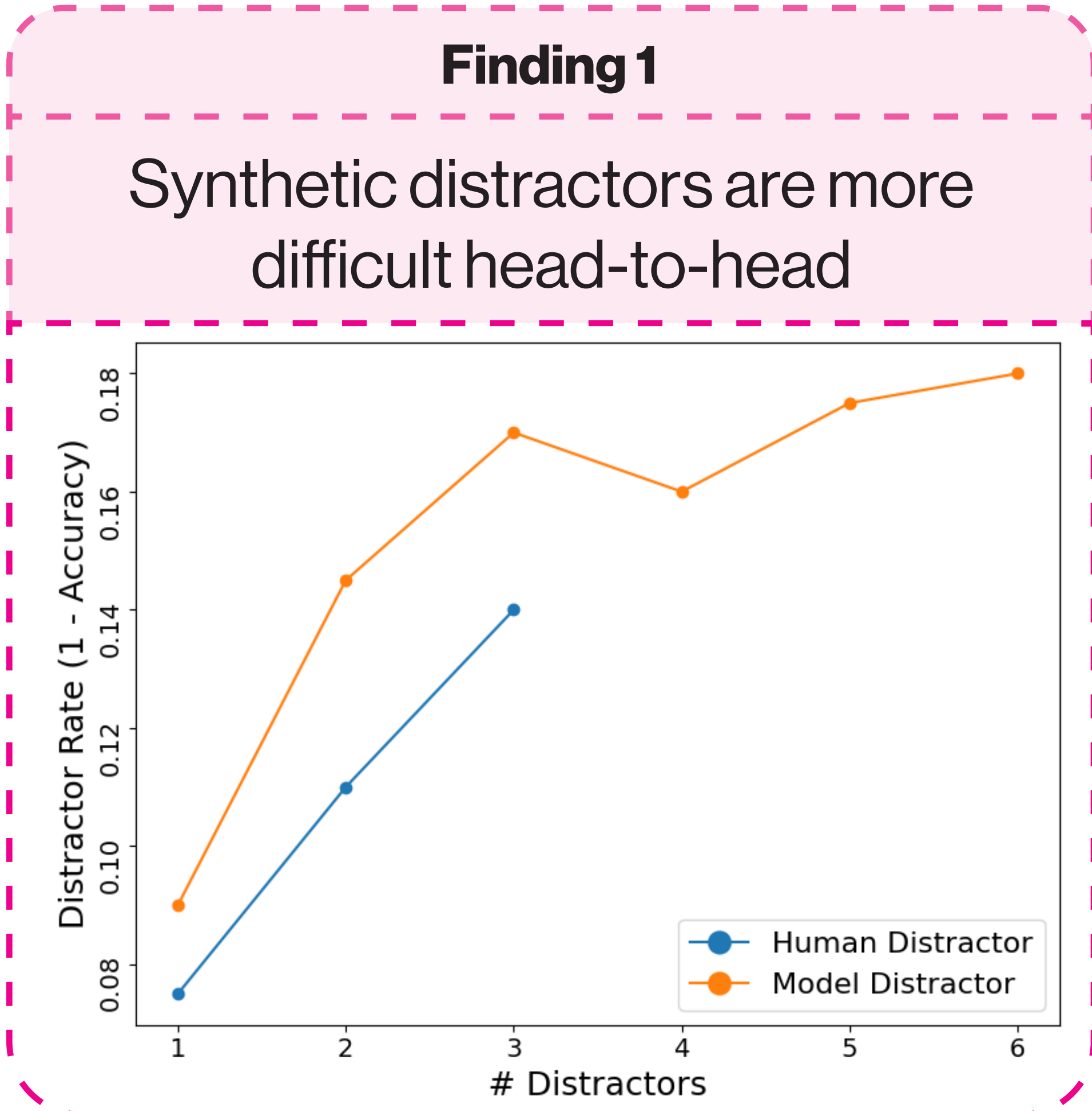


Figure 1: The synthetic distractor curve (orange) is consistently higher than human distractor curve (blue), indicating more

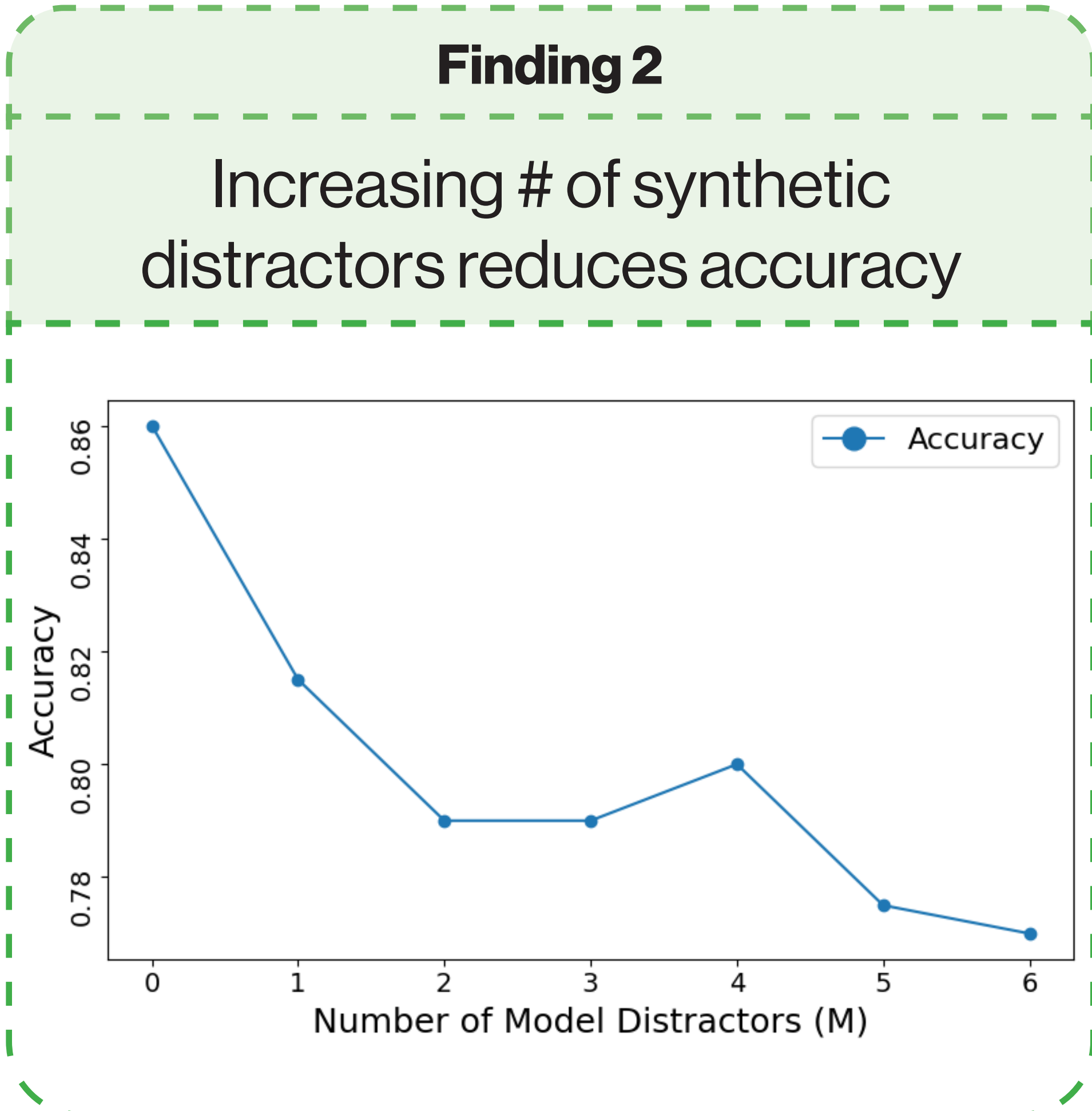


Figure 2: Model accuracy as LLM-generated distractors (M) are added to a base of 3 human distractors (H)

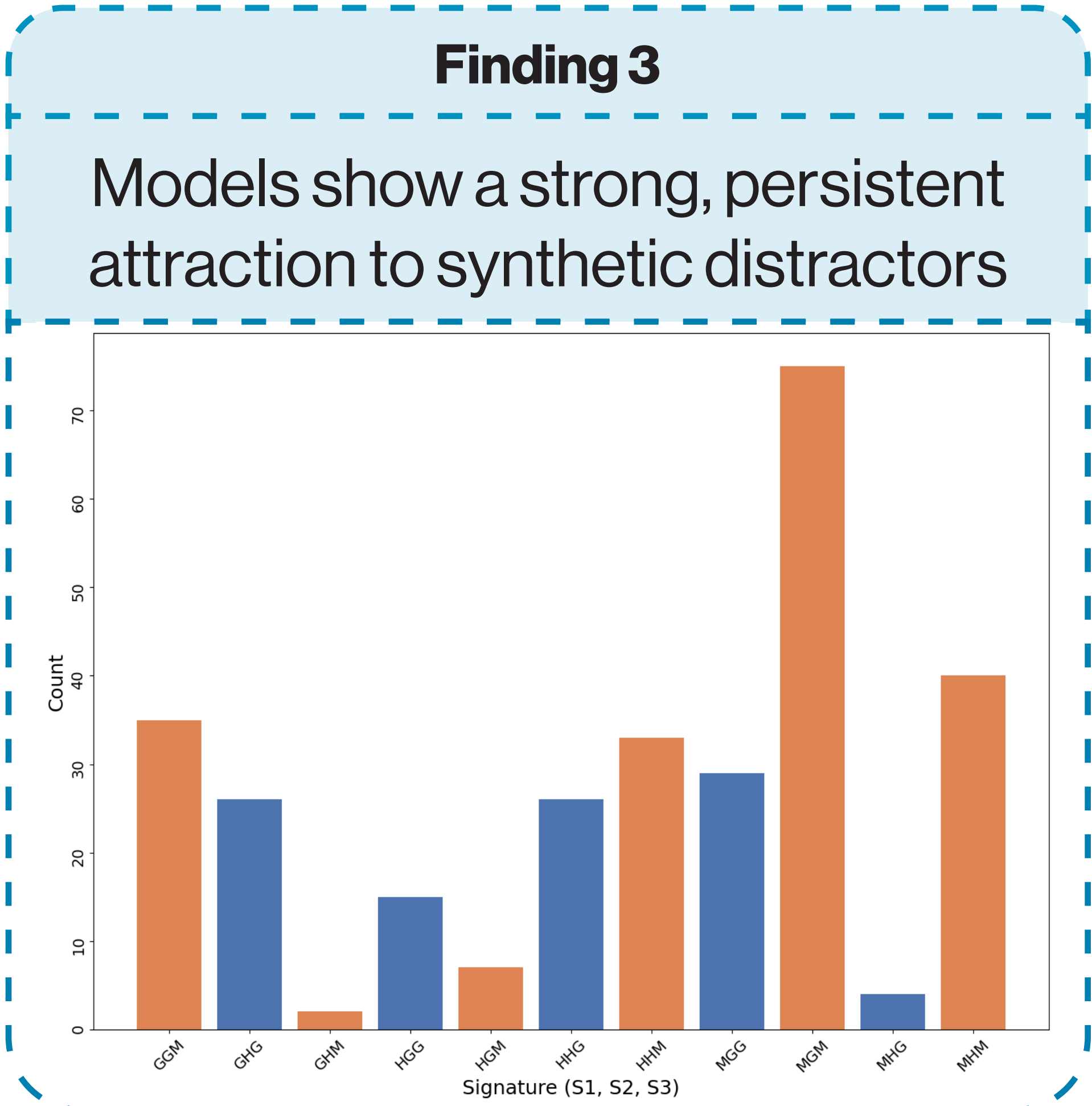


Figure 3: Frequencies of signatures, data for Signature GGG (70.4% of occurrences) is not shown to improve readability.

Analysis

Key Findings

- Synthetic distractors are disproportionately selected as answers, even when incorrect.
- Accuracy drops non-linearly when human and synthetic distractors are mixed, suggesting non-additive interference.
- Signature analysis reveals frequent prediction flips between settings

Future Work

- Quantify the benefit of conditioning on human distractors when generating synthetic distractors.
- Use sentence embeddings to measure if AI distractors are semantically “closer” to the correct answer.
- Replicate findings across model families and sizes.